

FINAL REPORT

DURIAN GRADING USING SIGNAL PROCESSING

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1.0 INTRODUCTION

Known as the "King of Fruits," durian is important both culturally and economically in Malaysia and other Southeast Asian nations. Musang King, the most expensive durian variety, is distinguished by its rich flavor, creamy texture, and high market value. Malaysia has experienced a boom in durian exports, notably frozen and vacuum-packed Musang King durians, due to rising demand from overseas, particularly from China, Singapore, and Hong Kong. Quality control in post-harvest procedures, especially grading, has therefore become more important. Each fruit is graded based on its ripeness, freshness, size, and general quality, which has an immediate effect on export credibility and consumer pleasure.

Traditionally, skilled personnel grade by hand using physical indicators such as size, colour, scent, and the sound made when the fruit is tapped. Even though these techniques have been employed for many years, they are labour-intensive, subjective, and inconsistent by nature. Depending on their own opinions, various employees may grade the same fruit differently, which could result in grading errors, decreased productivity, and the possibility of exporting inferior goods. Furthermore, since the interior quality cannot be confirmed once processed, frozen durians going to foreign markets need to be precisely assessed before opening. Additionally, the absence of digital tracking or documentation in manual techniques restricts supply chain transparency and traceability.

There is a great chance to use automation and artificial intelligence to revolutionize the grading process as smart farming and digital technologies become more widely used. An objective, real-time, and non-destructive grading system may be made possible by the combination of sensors, embedded electronics, AI-based image processing, and Internet of Things platforms. A Smart Durian Grading System created especially for Musang King durians is presented in this project. To categorize durians into quality ratings, the system makes use of machine learning, image analysis, and sensor data (presence, gas emissions). It is a useful and scalable solution for farmers, exporters, and packing stations looking to modernize their operations and maintain premium quality across all exports because it is based on the ESP32 microcontroller and connects to the Blynk IoT platform for remote data monitoring.

1.1 Problem Statement

Musang King durians have a high market value, but the grading system is still manual and antiquated, which makes it difficult to be consistent, accurate, and scalable. Because human grading is fundamentally subjective, various workers may have different opinions about the same durian, which could result in differing standards for quality. In the export industry, where a single shipment that is misgraded can sour relations with customers and ruin a brand's reputation, such discrepancies can be expensive. Furthermore, some durians are vacuum-sealed or frozen, which requires precise grading prior to fruit opening or processing. This calls for a non-invasive technique to assess internal ripeness and freshness, which is not possible with the manual methods used today. Additionally, traceability, transparency, and post-harvest decision-making are impacted by the absence of an automated record of the grading process due to the lack of digital integration. These problems point to a glaring weakness in the current system, the need for a smart, sensor-based system that can identify the physical and chemical characteristics of the fruit, categorize it using machine learning algorithms, and send the information to cloud platforms for tracking. Throughout the durian export value chain, closing these gaps can significantly increase operational effectiveness, lower human error, and guarantee greater customer satisfaction.

1.2 Objectives

This project's main goal is to create and deploy a Smart Durian Grading System that classifies Musang King durians according to quality using embedded electronics, AI-based image processing, and Internet of Things connection.

[1] To develop an automated detection system using an ultrasonic sensor that activates the grading process only when a durian is placed within a predefined distance from the sensor. This ensures the system operates efficiently by eliminating false triggers and reduces power consumption when no fruit is present.

[2] To detect signs of fermentation or spoilage in Musang King durians using an MQ-3 gas sensor, which measures alcohol vapours emitted by overripe or rotting fruit. This non-invasive method allows the system to identify fruits that are overripe or beginning to rot, which are not suitable for export.

[3] To classify the external quality of durians into Grade A or Grade B using image processing and a machine learning model trained via Edge Impulse and deployed on the ESP32-CAM module. The AI model evaluates key visual features such as size and colour to provide consistent and objective grading results.

[4] To implement real-time feedback and remote monitoring by displaying grading results on an OLED screen and uploading sensor data and classification output to the Blynk IoT platform. This enhances user interaction with the system and enables farm operators or exporters to access and monitor quality data from any location.

1.3 System Specification

The suggested system is based on the ESP32 microcontroller, which has a dual-core processor, built-in Wi-Fi, and support for several peripheral ports, making it ideal for this use case. Three main input sensors are used by the system. The grading procedure begins when an ultrasonic sensor identifies the presence of a durian in the system. Second, the air around the durian is monitored by a gas sensor (such the MQ-3) to identify chemicals like ethylene, which are organic byproducts of fruit fermentation and ripening. These measurements aid in identifying if the fruit is ripe, fresh, or starting to go bad. Third, the durian is photographed in high definition using an ESP32-CAM module. A pre-trained convolutional neural network (CNN) model from Edge Impulse is used to examine these photos. It uses learned visual patterns to categorize the fruit as Grade A or Grade B.

Three output components are part of the system. The user can see the classification result in real time on an OLED display. An LED indication, which is turned on during grading and off after completion, offers a straightforward and understandable status signal. Finally, sensor and grading data are transmitted via the ESP32's Wi-Fi connectivity to the Blynk IoT platform, where they are displayed on a dashboard with access to both historical and real-time data. The configuration is small and portable because every component is installed on a box. It can be used in rural farm sites or grading stations because power is provided by a USB or battery pack. The Edge Impulse firmware library, embedded C++ (Arduino IDE), and Blynk app integration are all part of the software design. All things considered, the system offers several benefits over manual grading techniques, including inexpensive deployment costs, real-time operation, and remote monitoring.

1.4 Scope of Engineering Project

In this engineering project, a non-destructive, AI-powered smart grading prototype created especially for Musang King durians will be developed, integrated, and tested. The system is designed for small-to-medium-sized businesses where reliable quality control is essential, like nearby farms, grading stations, and export packaging facilities. Sensor selection, microcontroller programming, AI model training and deployment, and IoT dashboard configuration are all included in the scope, which covers both hardware and software development. One durian at a time can be graded using the current prototype, which divides each fruit into three categories based on external appearance and environmental gas emissions: Grade A (premium), Grade B (standard), or Reject (spoiled or unripe). The project is limited by the specialization of the training dataset (which focuses on Musang King), camera image quality, and the limitations of embedded AI processing. It does not evaluate internal structure or weight at this stage, though such features can be incorporated in future iterations using weight sensors or acoustic testing. Future enhancements, like robotic arms for sorting, integration with packing lines, or grading of different durian kinds, can be added to the system with little alteration to the main platform because it is also made to be modular. In the end, this project shows how conventional agriculture may benefit from the application of contemporary engineering methods to create more intelligent, scalable, and data-driven solutions.

2.0 METHODOLOGY

This methodology details the structured approach, techniques, and resources used in developing a signal processing-based intelligent system for the purpose of automatic grading of durian. The system uses an ESP32 microcontroller equipped with built-in Wi-Fi and various port support to control three main types of sensors. The grading process begins with the detection of the presence of durians via an ultrasonic sensor, followed by ethylene gas monitoring using a gas sensor (MQ-3), and finally high-definition image analysis by the ESP32-CAM module supported by a pre-trained CNN model from Edge Impulse. The grading results are displayed in real-time via OLED and LED displays and transmitted to the Blynk platform for remote monitoring. This section will explain the design constraints, techniques used, resources involved, and the structure of the system block diagram and flowchart in detail

2.1 Design Constraints

In the development of the Durian Grading Using Signal Processing system, there are several technical constraints and challenges that have been identified throughout the design and implementation process. These constraints include aspects of software, hardware, as well as functional limitations of components used in real environments. Understanding and managing these constraints is very important to ensure that the developed system can function stably, accurately, and reliably.

First, the use of the Edge Impulse platform used to build AI models based on durian images faces constraints in terms of operating system compatibility. Edge Impulse produces EIM (Edge Impulse Model) models that need to be analysed through the AI Analyser in the Linux environment. Therefore, Windows users need to install the Windows Subsystem for Linux (WSL) first to ensure that the model can be analysed and run smoothly. Without WSL, AI functions cannot be fully utilised, and this limits the performance of the model during development.

Next, Visual Studio Code (VS Code) used as a code editing platform faces challenges in terms of port compatibility. When using ESP32 and ESP32-CAM camera, the communication port needs to be free and not shared by other devices as it can cause signal interference, and in some cases can cause the system to hang or crash. In addition, setting the delay timing in the

code is also very critical. If not adjusted properly, image transfer and communication between modules can be disrupted and affect the overall system performance.

Furthermore, the decision to change the use of ESP32-CAM to a regular USB Webcam was made because higher image quality and resolution are required for AI training purposes. However, this creates new constraints, where images from the webcam contain larger pixel sizes and need to be reprocessed via the resize function in OpenCV (Python) before being included in the AI model. The addition of this code increases the dependency on additional software and requires more complex processing.

Finally, in terms of AI data, challenges also exist when the number of image datasets used for training is still limited, especially since the system demo uses tomatoes as a substitute for real durians. As a result, the accuracy of the AI model decreases due to the lack of data representing various angles and real conditions. In addition, the use of the MQ-3 sensor which functions to detect volatile organic compounds (VOCs) could not be effectively demonstrated using tomatoes because tomatoes do not emit VOC gases like durian. Therefore, the sensor was only used as a proof of concept to show the potential for real applications on durian fruit in the future.

In conclusion, each design constraint identified presented a specific challenge to the project team. However, a deep understanding of each of these issues allowed the team to find alternatives and solve problems in a creative way. These constraints also helped to enhance the technical knowledge of the team members in the fields of signal processing, software engineering, and AI-based intelligent system integration.

2.2 Techniques Used

To make the system practical and portable to various locations, including rural farm sites, factories, or fruit grading stations, an important approach was to design a system that was compact and lightweight, yet still functionally robust. Therefore, one of the main techniques used was the design of printed circuit boards (PCBs) using Autodesk Eagle software. Eagle allows engineers to design their own connections between electronic components such as gas sensors, cameras, OLEDs, and ESP32s on a single, small board. This allows all components to be integrated into a special box, making it easier to install, store, and operate the system even in hard-to-reach areas.

This specially designed PCB board replaces the use of temporary and non-durable breadboards. It allows the system to be used repeatedly without the risk of disconnected or unstable connections. Eagle's design also helps to strategically arrange the components so that there is no electromagnetic interference and ensures that each component can function optimally. Figure 2.2.1 below shows the design made for the Printed Circuit Board (PCB) connection with a strategic arrangement using Autodesk Eagle software while figure 2.2.2 below shows the Printed Circuit Board (PCB) that has been completed using the Circuit Board Plotter at the Telephony Laboratory. In this project, the PCB layout is adapted to the shape of the box that houses the entire system, making it a fully portable device.

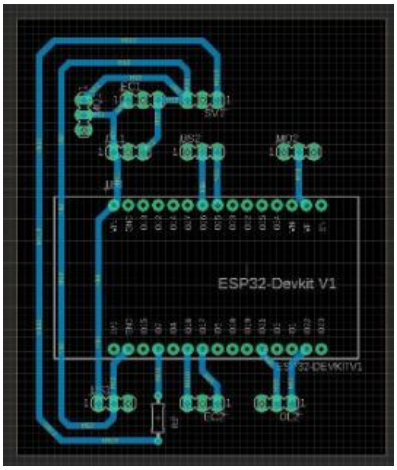


Figure 2.2.1: Design connection circuit for PCB using Autodesk Eagle software

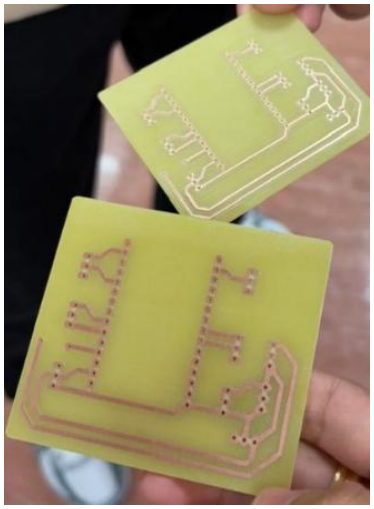


Figure 2.2.2: Printed Circuit Board (PCB) for this project.

In terms of gas sensing and analysis, the system uses an MQ3 sensor, which can detect the presence of volatile organic compounds (VOCs) such as ethylene gas released by durians as they ripen. The readings of these gases vary according to the ripeness of the durian – low for unripe fruit, medium for ripe fruit, and high for fruit that has started to spoil. A thresholding technique is used to divide durians into categories such as Unripe, Ripe, or Rotten. This allows the system to automatically provide a clear classification based on gas readings alone.

Next, image capture and colour analysis play an important role in improving the accuracy of the system. The ESP32-CAM module is used to automatically take pictures of the durian surface after detection. These images are analysed using an AI model based on a Convolutional Neural Network (CNN) through the Edge Impulse platform, which can detect colour patterns such as green, yellow, and brown, which are indicators of the stage of ripeness. Green usually indicates unripe durian, yellow as ripe, and brown as a sign of the beginning of spoilage.

Then, the system performs a combination of various input data, which is from gas sensors and image analysis. This technique allows the system to make intelligent decisions through a combination of AI, rather than relying on a single source of information. For example, if the gas reading shows an ambiguous threshold value, but the colour analysis shows a deep yellow, the system can more confidently identify that the durian is ripe. This process ensures that the grading results to determine grade A or B are more accurate, consistent, and reliable.

For data display and management, the system uses an OLED display to show durian grade information directly to the user after the process is complete. In addition, real-time data logging techniques are implemented - data from the sensors and grading results will be saved directly to an Excel file via a Wi-Fi connection and the use of the Blynk platform to monitor data remotely more easily by workers. With this, users or system administrators can view the history of durian grading that has been done previously and analyse the system's performance statistically.

Finally, the system uses LEDs as visual signals to indicate the status during operation. The LEDs will light up while the grading process is in progress and will turn off when the system has finished detecting and classifying the durian. This technique not only helps users quickly identify the system status without having to read the display but also helps in terms of operational safety so that users do not touch the system while it is active.

2.3 Resources Used

In the development of this system, the ESP32 microcontroller plays a key role as the 'brain' of the entire operation. The ESP32 was chosen for its compact yet high-power features, equipped with a dual-core processor and built-in Wi-Fi and Bluetooth connectivity. This allows the system to send sensor data and grading information to the Blynk IoT platform directly without the need for additional modules. Furthermore, the ESP32 supports a variety of digital and analogue connections, allowing it to communicate simultaneously with various sensors and external devices such as the MQ-3 sensor, the HC-SR04 ultrasonic sensor, OLED displays, and LED as shown in the Table 2.3 below. This capability makes the ESP32 an ideal choice for embedded system projects such as this one that require high reliability and integration.

Table 2.3: Hardware resources with their purpose

Resources Hardware	Purpose
ESP32 Wi-Fi + Bluetooth	Controls the entire system, including communication with sensors, OLED displays, LEDs, and Blynk IoT applications.
ESP32-CAM	Captures images of durian fruits for AI-based visual grading.
Ultrasonic Sensor (HC-SR04)	Detects the distance of durian for positioning as button before image capture.
MQ-3 Gas Sensor	Detects gas content (such as ethanol) related to durian ripeness.
OLED Visual Display	Displays results such as durian grade and system status.
LED Visual Indicator	Provides a visual indication of durian grade level.

In addition, the system also uses the ESP32-CAM, a version of the microcontroller that has a built-in camera to automatically capture images of the durian surface. This module can

capture images with sufficient resolution to conduct visual analysis based on machine learning (AI) using a CNN model. The captured images will be analysed to detect the durian's skin colour such as green, yellow, and brown, which are indicators of ripeness. The ESP32-CAM module can also be used to run AI models directly on the device through integration with the Edge Impulse platform, without the need for an external server. This allows the classification process to occur quickly and intelligently at the actual location.

Additional hardware components used include an MQ-3 sensor to detect the concentration of gases such as ethylene, which is used to determine the ripeness of durian. An HC-SR04 ultrasonic sensor is used to detect the physical presence of durian in the reading area. To provide direct information to the user, a 128x64 OLED display is used, which displays the grading results in real time. To indicate the operation of the system, LEDs are used as status indicators, which are lit when the grading process is in progress and go out when completed.

From a software perspective, various platforms have been used in the development process of this system. Edge Impulse is used to build and train a CNN-based AI model with the collected durian image data. This model is then uploaded to the ESP32-CAM for use in image classification directly. For developing more complex and manageable code, Visual Studio Code (VS Code) is used to manage project files and libraries more systematically. In addition, the system uses the Blynk IoT platform for real-time data display on a smartphone or monitor application. And uses Excel to record, store and analyse sensor data obtained during the testing process.



Figure 2.3.1: The Blynk software

Figure 2.3.1 above shows the Blynk software, an IoT platform that allows users to visually create mobile applications to monitor and control microcontroller devices such as ESP32. Among the main features of Blynk are a drag-and-drop interface for building application displays, the ability to receive data in real-time (real-time monitoring), as well as support for various widgets such as buttons, graphic displays, and value displays. In the Durian Grading

project, Blynk is used as the main medium to send and display the grading status of durian directly to the user's smartphone or monitor.



Figure 2.3.2: The Edge Impulse software

Figure 2.3.2 above shows the Edge Impulse software, a machine learning-based AI development platform for embedded devices. It provides a web-based interface for the data acquisition, training, and deployment of AI models. This platform supports the use of sensors such as microphones and cameras and allows model inference to be performed directly on microcontrollers such as ESP32. In this project, Edge Impulse is used to train a durian colour recognition model using data specification of durian to help the grading process be smarter.



Figure 2.3.3: The Visual Studio Code software

Next, Figure 2.3.3 above shows Visual Studio Code (VS Code), an open-source code editor that supports various programming languages such as C++, Python, and Arduino. It provides features such as syntax highlighting, IntelliSense, debugging, and various extensions including Platform IO and Arduino IDE. VS Code is used in this project as the main platform for writing, compiling, and uploading code to the ESP32 board that controls the entire grading system.



Figure 2.3.4: The Microsoft Excel software

Figure 2.3.4 above shows Microsoft Excel, a popular spreadsheet software for data analysis and management. It supports statistical, mathematical, data visualization, and filtering and sorting functions. Excel is used in this project to record and analyse sensor data and grading data by AI in real time.



Figure 2.3.5: The Autodesk EAGLE software

Finally, Figure 2.3.5 above shows Autodesk EAGLE software, an electronic circuit design software that allows users to design schematics and PCB layouts. It offers an extensive component library, two- and four-layer design tools, and integration with basic circuit simulation. In this project, EAGLE was used to professionally design the electronic component connections of the durian grading system before it was physically built on the PCB. Overall, this five software not only provide the necessary technical facilities but also speed up the system development process. The use of the right software helps in improving the accuracy of the durian grading system and facilitates the efficient integration process between hardware and software.

Finally, in terms of human resources, this project was developed by 5 groupmates Electronic Engineering majors who conducted collaborative research and system development. By leveraging knowledge in signal processing, microcontrollers, and embedded system programming to realize this system. This project was also guided by a supervising lecturer who provided guidance in terms of system design, component use, and field testing. In addition, online tutorials, official documentation from previous projects, Edge Impulse and technical forums were also used as important references to resolve various technical issues throughout the system development. This combination of human resources and technology has enabled the creation of a smart, practical and scalable durian grading system for the industry.

2.4 Block Diagram

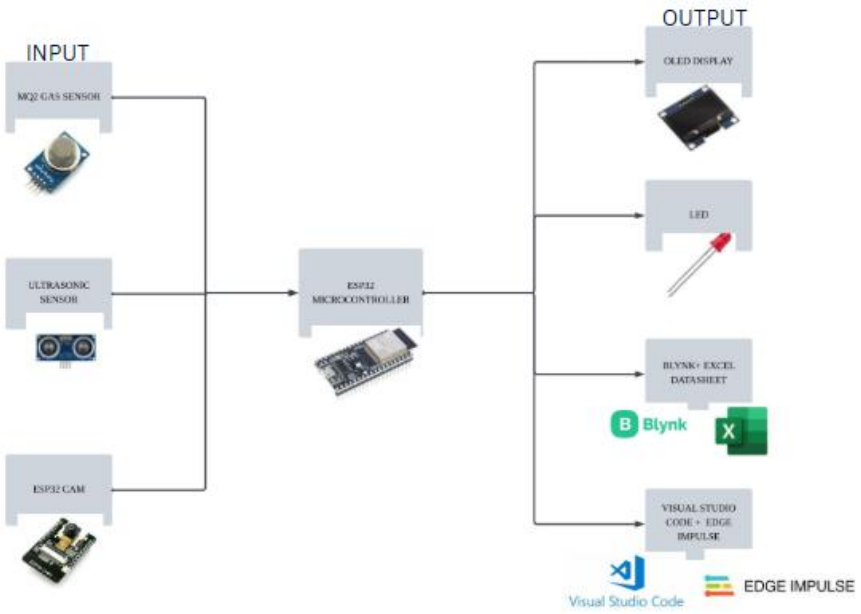


Figure 2.4.1: The Block Diagram of The System

Figure 2.4.1 above shows the flow of the Durian Grading System using ESP32 as the main microcontroller that connects the input and output sections. In the input section, there are three main components, namely MQ3 Gas Sensor, Ultrasonic Sensor, and ESP32-CAM. MQ3 Gas Sensor is used to detect the presence of gases such as methane, ethylene or alcohol that may be related to the ripeness of durian. Ultrasonic Sensor measures the distance between the sensor and the object, which is durian, helping to determine the presence of durian fruit. ESP32-CAM is used to take images of durian fruit as data for visual assessment.

Next, ESP32 Microcontroller acts as a control centre that receives input from all three sensors. It processes the data received and makes decisions based on the trained model. This ESP32 also has built-in Wi-Fi capabilities that allow it to send information wirelessly to other platforms such as Blynk and Microsoft Excel. In addition, this ESP32 has been programmed using Visual Studio Code software with the help of Edge Impulse machine learning platform to conduct inference based on AI models.

For the output section, there are several main components, namely OLED Display, LED, Blynk and Microsoft Excel, as well as Visual Studio Code and Edge Impulse. The OLED Display is used to display results such as the category of durian detected, which is grade A or B. The LED is used as a quick visual indicator, which is signals to indicate the status during

operation. Data is also sent to the Blynk application and recorded in Excel for monitoring and analysis purposes. Visual Studio Code and Edge Impulse, although not functioning as physical components, play an important role in the development and training of the AI model used in this system.

2.5 Flowchart

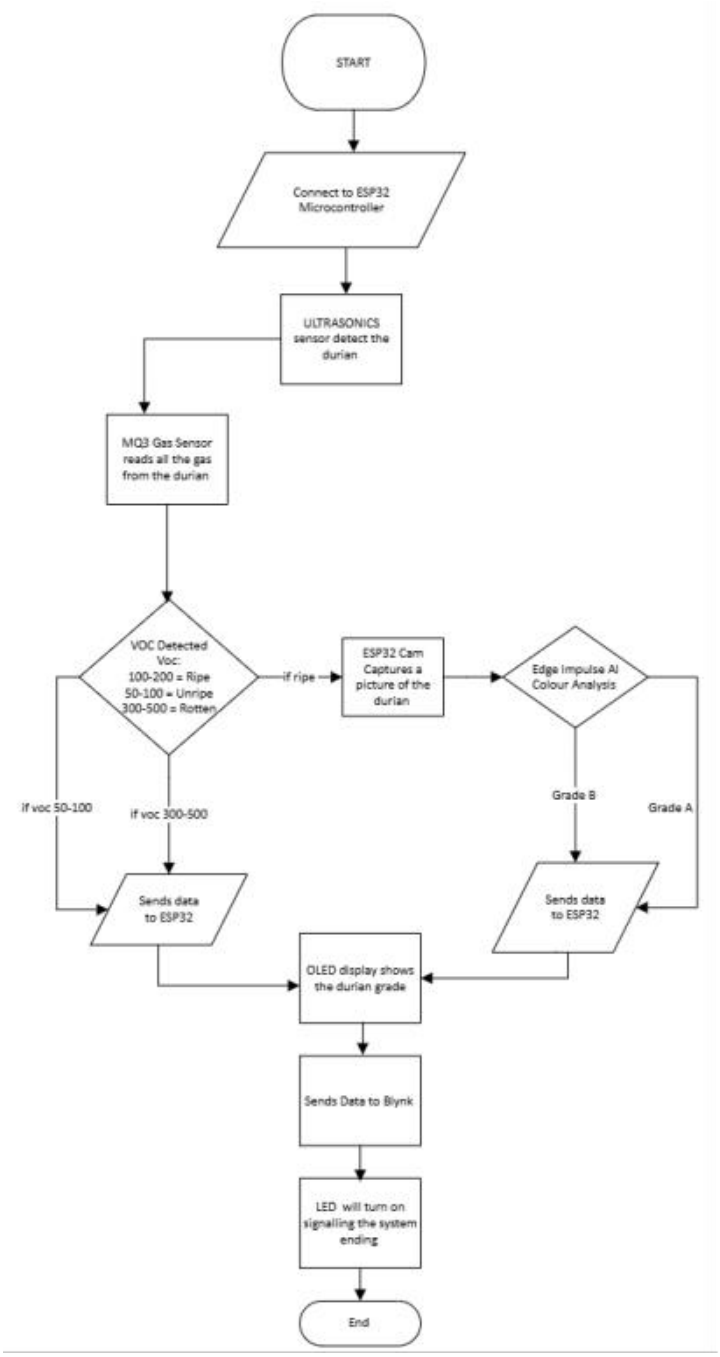


Figure 2.5.1: The Flowchart of The System

Figure 2.5.1 above shows a complete flowchart for the operational process of the developed automatic durian grading system. This process involves several main stages that are logically and systematically interconnected, with the use of various types of electronic components and software technology to achieve automatic, accurate and efficient grading of durian fruit. The system starts with the activation of the ESP32 Microcontroller, which acts as the main processing and control centre of the entire system. The ESP32 plays an important role in receiving input from various sensors and sending output to the display and communication modules.

The first step in the process is the detection of the presence of an object, which is the durian fruit, using an ultrasonic sensor. This sensor will measure the distance in front of it and if there is an object detected within a certain range, the system will assume that a durian has been placed in the right place. Next, the system activates the MQ3 gas sensor which functions to detect the concentration of volatile organic compounds (VOC) released by the durian fruit. This sensor reads the VOC concentration value in ppm (part per million) units. This gas is important because it is closely related to the maturity level of the durian fruit.

The gas readings obtained are used as a basis for determining the ripeness of the durian:

1. If the reading is between 50–100 ppm, the durian is categorized as unripe.

2. If the reading is between 100–200 ppm, the durian is classified as ripe.

3. If the reading is between 200–500 ppm, the durian is considered rotten due to excessive gas emission.

For durians identified as ripe, the system will activate the ESP32-CAM, which is an ESP32-based camera module that is capable of recording and transmitting images directly. This camera will take a picture of the durian and send it to an artificial intelligence (AI) model developed through the Edge Impulse platform using VS code. This AI model will conduct visual analysis based on the surface colour characteristics of the durian to determine the fruit grade, whether Grade A or Grade B. This AI inference process is performed using Visual Studio Code (VS Code) which operates the trained model from Edge Impulse and facilitates communication between the camera and the microcontroller.

The output or results from the AI analysis will then be sent back to the ESP32 to be displayed to the user. This final durian grade information will be:

1. Displayed in real time via the OLED Display, so that the user can easily see the results.

2. Sent to the Blynk mobile application via Wi-Fi connection, allowing the user to monitor the data via smartphone or PC.

3. Automatically recorded into an Excel file, providing a historical record for further analysis or documentation.

As a final step, the system will light up an LED as a visual indicator that the entire process has been completed. This LED light signal that the user can remove the durian, and the system is ready for the next grading session.

With this careful arrangement of operations, the automatic durian grading system not only speeds up the process but also increases the accuracy, consistency and reliability in determining the quality of durian objectively and systematically. It also provides remote monitoring facilities and data collection that can be used for reporting and process improvements in the future.

3.0 RESULT AND DISCUSSION

This section of the report focuses at the overall performance and output of the automatic durian grading system during the demonstration. A tomato was employed as a replacement for a durian during testing. The technology combines sensor-based detection and AI-based image processing to assess fruit quality in a non-invasive and automated way based on gas emissions and visual attributes using integrated sensors and intelligent processing. The discussion is separated into two major sections, which include hardware implementation and software integration, with a comprehensive examination of the results collected from the sensors and AI model during the testing.

3.1 Hardware

The automatic durian grading system's hardware focuses on the integration of several electrical components that allow for sensory detection, data collection, and output display. Each hardware part is chosen and designed to play a specific role in the grading process, allowing the system to operate in a logical and automated manner. This part describes the operation of each hardware component, discusses how they interact in order according to the designated system flowchart, and displays the data and results obtained when testing with a tomato as a substitute for a durian.



Figure 1 Tomato detection by ultrasonic sensor

Based on Figure 1, the ultrasonic sensor was placed above the fruit placement location to detect it's presence before initiating the VOC reading and grading procedure. The system was

calibrated to detect any item within a 15 cm range. Once the tomato was positioned beneath the sensor and within this distance threshold, the system identified the fruit's presence and moved on to the next phase. This technique guarantees that the grading system is activated only when a fruit is physically present, preventing insignificant readings or processing.



Figure 2 VOC measurement done by MQ3 sensor

The MQ3 gas sensor detects the quantity of volatile organic compounds (VOCs) released, which is critical in determining fruit ripeness. The real sensor output is read as an analog value ranging from 0 to 4095, which corresponds to the ESP32 microcontroller's ADC resolution. After detecting the item, the system analyses the VOC level for 15 seconds shown in Figure 2 before providing the results. This delay is deliberate and necessary for the MQ3 sensor to stabilize and precisely detect the gas concentration. The warm-up or stabilization time reduces noise and inaccuracy caused by sudden variations in the surrounding air.

As shown in the Figure 2, the values obtained are quite high which is around 1707. While this is not an exact VOC concentration in ppm, it does demonstrate that the tomato emitted a measurable quantity of gas, imitating a real-world scenario in which a durian releases fermentation gases when ripening. This number was accurately presented on the OLED screen as part of the grading result, demonstrating the sensor's responsiveness and integration into the system workflow.

For the purpose of demonstration, the VOC output logic in the code was modified to emulate ripeness detection. This choice occurred because, unlike durians, tomatoes produce very little or no noticeable fermentation gasses under typical room circumstances. As a result, to test the system's VOC detection and grading procedure, we modified the code to interpret the analog value (e.g., 1712) as a "ripe" state. This enabled the system to go on to the AI-based grading stage, guaranteeing that the entire grading cycle could be evaluated and confirmed with a substitute fruit.

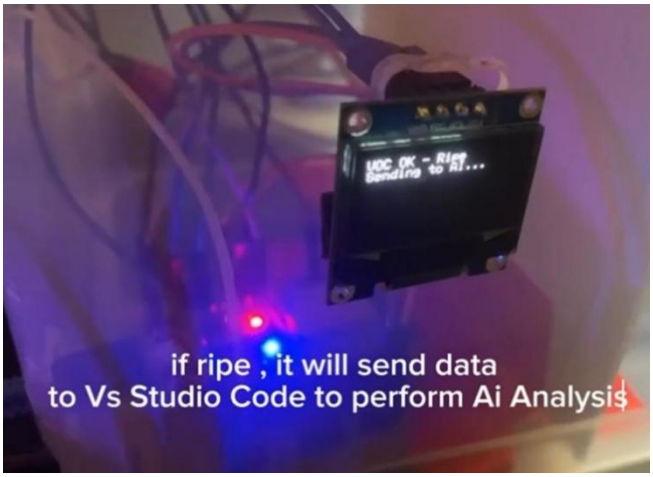


Figure 3 starting for Image based grading using AI

Initially, the system was intended to use the ESP32-CAM to capture images for grading. However, due to constraints such as inconsistent picture capture, low resolution, and trouble calibrating illumination, the team moved to a normal USB webcam attached to a computer for image acquisition. Shown in Figure 3, the captured image was analysed by Visual Studio Code that is attached with the EIM file that contains AI analysis based on the data base from Edge Impulse's which is a pre-trained AI model, that will then classifies the fruit as Grade A or Grade B based on visual characteristics such as color consistency and shape.

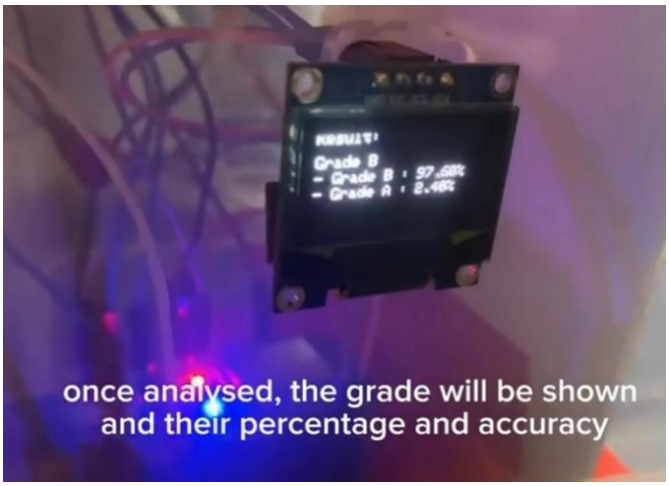


Figure 4 AI analysis output

This result was then relayed back to the ESP32 and shown on the OLED panel, along with the the accuracy of the fruit's grade offering an instant and comprehensive overview of the analysis results. By providing the classification confidence about the fruit's grade obtained in Figure 4 where the tomato is analysed as Grade B with an accuracy of 97.68%, it improves the system's transparency and resilience, which is especially beneficial in practical implementation.

3.2 Software

The software component of the automatic durian grading system is important to allowing real-time data processing, intelligent categorization, and remote monitoring. While numerous sensors and imaging equipment give raw inputs, the software processes, analyzes, and delivers the data in a useful manner. The system's software is intended to function across three major platforms such as Edge Impulse for AI model training and inference, Visual Studio Code (VS Code) for running classification scripts from the exported EIM file, and the Blynk IoT platform for remote data monitoring. The solution additionally enables to log outcomes to Excel spreadsheets, which helps with data traceability and performance evaluation.

This section describes how these tools interact with the hardware system, how they contribute to overall system operation, and the results obtained after testing with a tomato as the subject fruit. The combination of intelligent decision-making with IoT-based visualization illustrates the feasibility of a scalable, low-cost smart agricultural system.

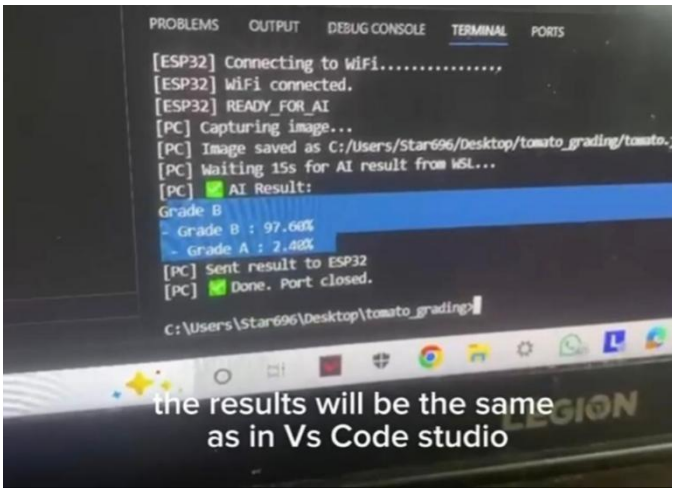


Figure 5 Terminal output shown in Visual Studio Code
sync well with the output on OLED

Based in Figure 5, shows the terminal output from Visual Studio Code (VS Code) during the AI-based image analysis process. This step is a key part of the software pipeline, where the captured image of the tomato is classified into a grade based on visual features using the trained Edge Impulse model (EIM file). A 15-second delay is used to allow the AI model (running in a local environment like Windows Subsystem for Linux, or WSL) to load, process the picture, and produce an inference result. This delay ensures that the output is precise and reliable.

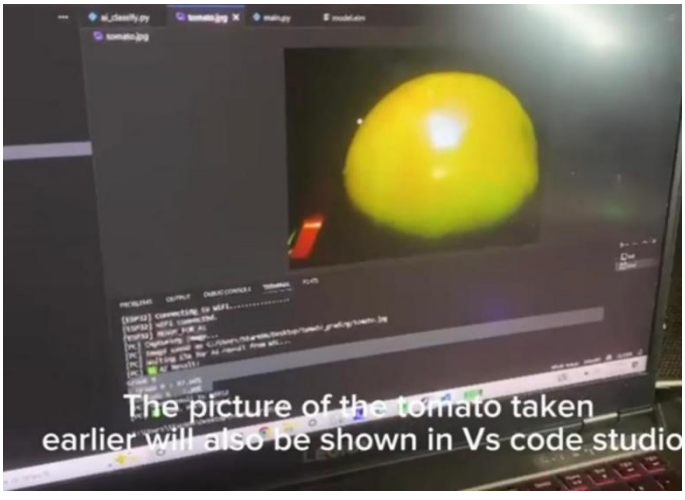


Figure 6 Webcam captures the tomato and displays on
Visual Studio Code

In Figure 6, the image above shows the program execution step, in which the captured tomato image is presented in the VS Code environment. The software uses Python to import and display the image as part of a user-interactive categorization pipeline. This stage ensures that picture capture, loading, and GUI response have been successfully integrated, which is critical for system usability and debugging.

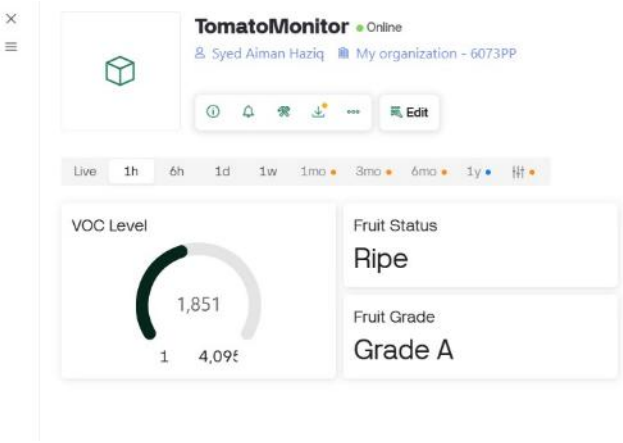


Figure 7 Blynk web dashboard for real time monitoring

The integration of the Blynk IoT platform is critical for real-time monitoring, remote access, and data visualization in the automatic durian grading system. In this project, after the ESP32 gets the fruit's VOC level and image-based grade, it sends the data to the Blynk Cloud, where it is accessible via the Blynk mobile app.

Based on Figure 7, the Blynk web dashboard that has been set up, displays the VOC level detected by the MQ3 sensor with a value of 1851. The fruit status obtained from the OLED display shown ripe due to its value of VOC level that has been modified in the program. While the fruit grade result is obtained from the AI image analysis by the webcam. This enables users to monitor the system from anywhere, without having to be physically present near the hardware. It is particularly beneficial in commercial applications where various grading systems may be functioning in separate locations.

	A	B	C	D	E	F
25	24	1706	Rotten	-	7/9/2025 16:27:22	
26	25	1821	Rotten	-	7/9/2025 16:27:41	
27	26	1820	Unripe	-	7/9/2025 16:28:29	
28	27	1824	Unripe	-	7/9/2025 16:28:46	
29	28	1817	Unripe	-	7/9/2025 16:29:02	
30	29	1831	Unripe	-	7/9/2025 16:29:18	
31	30	1843	Unripe	-	7/9/2025 16:29:37	
32	31	1846	Unripe	-	7/9/2025 16:29:54	
33	32	1842	Unripe	-	7/9/2025 16:30:10	
34	33	1914	Ripe	Grade B	7/9/2025 16:33:53	
35	34	1933	Ripe	Grade A	7/9/2025 16:39:24	
36						
37						

Figure 7 Excel datasheet for data logging

To aid in data analysis and performance monitoring, the system offers a feature that logs data to an Excel spreadsheet shown in Figure 7, including the date and time, VOC level, fruit state, and fruit grade. This tool promotes traceability by allowing users to follow trends over time, do statistical analysis using graphs and charts, and evaluate the quality consistency of various fruits.

4.0 CONCLUSION

Using embedded technologies, artificial intelligence (AI), and the Internet of Things, the Smart Durian Grading System successfully accomplished its primary goals of automating the grading procedure for Musang King durians. In order to ensure that the system only ran when required, an ultrasonic sensor was integrated into the system to detect the presence of durians and initiate the grading sequence automatically. The MQ-3 gas sensor was successful in identifying overripe or damaged durians by detecting the alcohol-based vapors created during fermentation, despite its inability to directly detect ethylene. As a result, fruits that were unfit for export could be filtered out of the system. In the meantime, the ESP32-CAM module effectively recorded and categorized the durians' external characteristics using a machine learning model trained with Edge Impulse. This allowed for the determination of quality grades (A or B) based on visual criteria like size and colour.

The outputs of the system also performed as planned. The LED indication visually indicated when a fruit was being processed, and the OLED display gave instant on-site feedback by displaying the grading result in real time. Crucially, all sensor data, grading results, and timestamps were transmitted wirelessly for remote monitoring and digital documentation through the usage of the ESP32's Wi-Fi capabilities and the Blynk IoT platform. Improved traceability and data-driven decision-making for farm operators or exporters were made possible by this real-time connectivity, which was in line with contemporary agricultural requirements. The effective fusion of hardware, software, and artificial intelligence components showed that readily accessible electronics might be used to create a portable, low-cost smart grading system.

Overall, the project's findings confirm that an automated, non-destructive durian grading system is both practical and advantageous, particularly for premium kinds like Musang King. In addition to reducing the need for manual labour, the technology improved grading consistency and added digital transparency to the post-harvest process. The current prototype provides a strong basis for practical applications, even though potential future advancements could include robotic handling, internal structural analysis, or sensors tailored to ethylene. It offers a scalable solution that can bolster Malaysia's competitive advantage in the international durian export market and encourage agricultural innovation.

6.0 REFERENCES

[1] S. Ditcharoen, P. Sirisomboon, K. Saengprachatanarug, A. Phuphaphud, R. Rittiron, A. Terdwongworakul, C. Malai, C. Saenphon, L. Panduangnate, and J. Posom, "Improving the non-destructive maturity classification model for durian fruit using near-infrared spectroscopy," Postharvest Biology and Technology, vol. 190, p. 111949, 2022. doi: 10.1016/j.postharvbio.2022.111949.

[2] D. Diana, T. B. Kurniawan, D. A. Dewi, M. K. Alqudah, M. K. Alqudah, M. Z. Zakari, and E. F. B. E. Fuad, "Convolutional neural network based deep learning model for accurate classification of durian types," Journal of Applied Data Sciences, vol. 6, no. 1, pp. 101–114, Jan. 2025.

[3] A. M. Bratu, M. Petrus, and C. Popa, "Monitoring of post-harvest maturation processes inside stored fruit using photoacoustic gas sensing spectroscopy," Sensors, vol. 20, no. 12, pp. 1–14, Jun. 2020.

[4] F. S. A. Sa'ad, M. F. Ibrahim, A. Y. Md. Shakaff, A. Zakaria, and M. Z. Abdullah, "Shape and weight grading of mangoes using visible imaging," Postharvest Biology and Technology, vol. 62, no. 2, pp. 171–179, Nov. 2011.

[5] M.-H. Nguyen, N.-V. Pham, T.-Q. Luu, and D.-D. Ngo, "Artificial intelligence-based non-invasive durian classification uses knocking sounds and vibration signals," in Proc. 2024 7th Int. Conf. Green Technology and Sustainable Development (GTSD), Ho Chi Minh City, Vietnam, 2024, pp. 259–264, doi: 10.1109/GTSD62346.2024.10674976.

[6] M. Rivaia, F. Budiman, D. Purwanto, M. S. A. Al Baida, T. Tukadi, and D. Aulia, "Discrimination of durian ripeness level using gas sensors and neural network," in Proc. Sixth Information Systems International Conference (ISICO 2021), Surabaya, Indonesia, 2021, pp. 1–6.

[7] S. Radi, W. S. Ciptohadijoyo, W. S. Litananda, M. Rivai, and M. H. Purnomo, "Electronic nose based on partition column integrated with gas sensor for fruit identification and classification," Comput. Electron. Agricult., vol. 121, pp. 429–435, 2016, doi: 10.1016/j.compag.2015.11.013.

7.0 TIKTOK POSTING

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